CAUSALITY ASSESSMENT

Professional & Systematic Pharmacovigilance Notes

Video reference	“Causality Assessment” — requested YouTube video (video ID: 6gH4AWnwTDI)
Core framework	WHO–UMC standardized case causality assessment; PvPI/IPC guidance; Naranjo ADR Probability Scale for comparison
Best suited for	Pharmacovigilance students, Pharm.D/clinical pharmacy learners, ADR monitoring and drug-safety interview preparation

Important: Causality assessment estimates the likelihood of a relationship; it does not prove that a medicine caused an event. The WHO notes that spontaneous ADR reports commonly contain uncertainty and that causality methods classify relationship likelihood rather than provide a precise quantitative probability.

1. What is Causality Assessment?

Causality assessment is the structured evaluation of whether there is a plausible relationship between a suspected medicinal product and an adverse event/reaction. It considers the clinical-pharmacological history and the quality/completeness of the case documentation.

- It is performed on an individual case report to classify the strength of the drug–event relationship.
- It considers the timing of drug exposure and event, alternative explanations, dechallenge, and—when available and ethically appropriate—rechallenge.
- Most spontaneous reports cannot establish causality with absolute certainty because diagnostic tests may be unavailable and rechallenge is often not ethically justified.
- The WHO–UMC system is intended as a practical, standardized approach for case reports.

2. Why is Causality Assessment Important?

- Helps characterize individual safety reports consistently.
- Reduces disagreement between assessors by providing a common framework.
- Supports signal detection and aggregate safety evaluation.
- Helps distinguish cases with stronger evidence of a drug relationship from those with major uncertainty or competing explanations.
- Provides an educational and scientific framework for reviewing case histories.

What it cannot do

- It cannot prove a drug caused an event in the same way a controlled experiment can.
- It cannot produce an exact numerical probability that a drug caused an event.
- It cannot convert incomplete or contradictory documentation into certainty.
- It does not replace clinical diagnosis or clinical judgment.

3. Key Evidence Used in Causality Assessment

- Temporal relationship: Did the event occur within a biologically/pharmacologically plausible period after exposure?
- Alternative causes: Could the disease, another medicine, interaction, infection, comorbidity or another factor explain the event?
- Dechallenge: What happened after withdrawal or reduction of the suspected drug?
- Rechallenge: Did the event recur after re-exposure? This may provide strong evidence but should never be performed solely to establish causality when it would pose unacceptable risk.
- Known pharmacology/phenomenology: Is the event compatible with the pharmacological action or a recognized reaction pattern?
- Objective evidence: Are there laboratory, imaging, clinical or other findings supporting the event?
- Dose-response: Did the event change with dose where such information is clinically meaningful?
- Previous evidence: Are similar reactions known with the drug or a related agent?

4. WHO–UMC Causality Assessment Scale

PvPI uses the WHO–UMC causality assessment system. It has six standard categories: Certain, Probable/Likely, Possible, Unlikely, Conditional/Unclassified, and Unassessable/Unclassifiable.

Category	Core interpretation	Key features
Certain	Strong causal relationship	Plausible timing; no reasonable alternative explanation; plausible response to withdrawal; event is definitive pharmacologically/phenomenologically; satisfactory rechallenge if necessary.
Probable / Likely	Relationship is clinically reasonable and alternatives are unlikely	Reasonable timing; unlikely to be explained by disease/other drugs; clinically reasonable response to withdrawal; rechallenge not required.
Possible	Drug may be related, but alternatives exist	Reasonable timing; could also be explained by disease or other drugs; withdrawal information may be lacking or unclear.
Unlikely	Relationship is improbable	Timing makes relationship improbable, although not impossible; disease or another drug provides a plausible explanation.
Conditional / Unclassified	Insufficient for final classification but more information may resolve it	Event/laboratory abnormality is present, but more data are needed or additional data are under examination.
Unassessable / Unclassifiable	Cannot be judged	Report suggests an ADR, but information is insufficient/contradictory and cannot be supplemented or verified.

Memory aid: C → P → Po → U → C/U → U/U = Certain → Probable/Likely → Possible → Unlikely → Conditional/Unclassified → Unassessable/Unclassifiable.

5. Understanding Each WHO–UMC Category

Certain

The strongest category. The chronology is plausible, alternative explanations are absent, dechallenge is plausible, and the event is definitive pharmacologically or phenomenologically. Rechallenge may be relevant when necessary and satisfactory.

Probable / Likely

There is a reasonable time relationship, competing explanations are unlikely, and withdrawal produces a clinically reasonable response. Rechallenge is not required.

Possible

Timing is reasonable, but the event can also be explained by disease or another medicine. Dechallenge information may be missing or unclear.

Unlikely

The timing makes a drug relationship improbable and another explanation is plausible. The relationship is not impossible.

Conditional / Unclassified

There is an event or laboratory abnormality, but more information is needed for proper assessment or additional information is still being examined.

Unassessable / Unclassifiable

The report suggests an adverse reaction, but the information is insufficient or contradictory and cannot be supplemented or verified.

6. Dechallenge vs Rechallenge

Feature	Dechallenge	Rechallenge
Meaning	What happens after stopping/reducing the suspected medicine	What happens after the suspected medicine is restarted
Evidence	Improvement after withdrawal can support causality	Recurrence after re-exposure can strongly support causality
Requirement	Often considered in causality assessment	Not required for probable/likely; may be relevant for certain
Safety	Must be interpreted with the natural course of disease	Should not be deliberately performed when medically unsafe or unethical

Important distinction: absence of a dechallenge response does not automatically rule out causality; the event may be irreversible, self-limited, or influenced by disease course and other factors.

7. Practical Causality Assessment Workflow

- Step 1 — Define the adverse event clearly.
- Step 2 — Establish the complete medication timeline: start date, dose, route, frequency, changes and stop date.
- Step 3 — Establish the event timeline: onset, progression, treatment and outcome.
- Step 4 — Assess whether the time-to-onset is plausible for the suspected medicine.
- Step 5 — Search for alternative explanations: underlying disease, concomitant medicines, interactions, infections, comorbidities and other exposures.
- Step 6 — Review dechallenge information.
- Step 7 — Review rechallenge information only if it occurred clinically.
- Step 8 — Consider pharmacological/clinical plausibility and objective evidence.
- Step 9 — Select the WHO–UMC category that best fits the total evidence.
- Step 10 — Document missing information and consider follow-up where appropriate.

8. Naranjo ADR Probability Scale — Comparison

The Naranjo algorithm is a questionnaire-based ADR probability scale. It is useful for learning and comparison with WHO–UMC, but it is not the same as the WHO–UMC qualitative system.

Score	Interpretation
≥ 9	Definite
5–8	Probable
1–4	Possible
≤ 0	Doubtful

The 10 Naranjo questions cover previous reports, temporal sequence, dechallenge, rechallenge, alternative causes, placebo response, toxic drug concentration, dose-response, previous similar reaction, and objective evidence.

9. Naranjo Questions — Exam Revision

1. Are there previous conclusive reports on this reaction?
2. Did the adverse event occur after administration of the suspected drug?
3. Did the reaction improve when the drug was discontinued or an antagonist was given?
4. Did the reaction reappear after readministration?
5. Are there alternative causes that could independently explain the reaction?
6. Did the reaction reappear with placebo?
7. Was the drug detected in blood/body fluids at a toxic concentration?
8. Did the reaction become more severe with dose increase or less severe with dose decrease?
9. Did the patient have a similar reaction to the same or related drug previously?
10. Was the reaction confirmed by objective evidence?

10. WHO–UMC vs Naranjo — Quick Comparison

Aspect	WHO–UMC	Naranjo
Approach	Qualitative clinical judgment framework	Structured point-based questionnaire
Output	Six categories	Score with probability category
Main emphasis	Clinical-pharmacological relationship + documentation quality	Ten predefined questions with weighted points
Used in PvPI	Yes — PvPI uses WHO–UMC causality assessment	Useful for comparison/education; not the PvPI WHO–UMC framework
Strength	Flexible clinical assessment for spontaneous reports	Simple, reproducible scoring structure

Exam point: Do not interchange WHO-UMC categories with Naranjo scores. They are different assessment approaches.

11. Worked Example — Drug-Associated Rash

A patient begins Drug A. Two days later a generalized rash develops. Drug A is stopped. The rash improves over the next several days. No new medicines are identified and no infectious cause is documented.

- Temporal relationship: reasonable.
- Alternative causes: none clearly documented.
- Dechallenge: clinically reasonable improvement after withdrawal.
- Rechallenge: not performed.
- Overall: the case may fit Probable/Likely under WHO–UMC if the full clinical documentation supports these conclusions.

The example demonstrates reasoning, not an automatic classification rule. The final assessment depends on the complete case history and quality of evidence.

12. Common Errors in Causality Assessment

- Assuming that temporal association alone proves causality.
- Ignoring alternative causes such as underlying disease or concomitant medicines.
- Treating dechallenge as definitive proof without considering the natural course of the illness.
- Assuming rechallenge is necessary for every case; it is not.
- Performing or recommending intentional rechallenge solely for causality assessment when it could harm the patient.
- Choosing a higher causality category because the reaction is severe or serious. Seriousness and causality are different concepts.
- Using a Naranjo score when the required information is unavailable and then presenting the result as certain.
- Ignoring contradictory or unverifiable information; such cases may fall under unassessable/unclassifiable.

13. Causality ≠ Severity ≠ Seriousness ≠ Expectedness

Term	Meaning
Causality	How likely the medicine is related to the event.
Severity	Intensity of the event (e.g., mild, moderate, severe).
Seriousness	Regulatory classification based on outcomes/criteria such as death, life-threatening event, hospitalization, disability or congenital anomaly.
Expectedness	Whether the nature/severity/specificity of the reaction is consistent with existing product safety information.

This distinction is frequently tested in pharmacovigilance examinations and interviews.

14. One-Page Revision Algorithm

ADR/event identified

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Build drug + event timeline

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Check temporal plausibility

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Look for alternative causes

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Assess dechallenge

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Review any actual rechallenge

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Assess pharmacological/clinical plausibility + objective evidence

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Select WHO–UMC category

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Document reasoning + missing information + follow-up needs

15. Interview / Viva Questions

Q. What is causality assessment?

A. A structured evaluation of the likelihood of a relationship between a medicinal product and an adverse event/reaction.

Q. Which causality system is used by PvPI?

A. WHO–UMC causality assessment system.

Q. How many WHO–UMC categories are there?

A. Six.

Q. What are they?

A. Certain; Probable/Likely; Possible; Unlikely; Conditional/Unclassified; Unassessable/Unclassifiable.

Q. Is rechallenge mandatory for Probable/Likely?

A. No.

Q. What does Possible mean?

A. There is a reasonable time relationship, but the event can also be explained by disease or another drug; withdrawal information may be lacking or unclear.

Q. What is Unassessable/Unclassifiable?

A. A suspected ADR cannot be judged because information is insufficient or contradictory and cannot be supplemented or verified.

Q. Does causality assessment prove causation?

A. No. It classifies the likelihood of a relationship based on available evidence.

Q. What is Naranjo?

A. A structured ADR probability scale based on ten questions and a weighted score.

16. Key Takeaways

- Causality assessment is a structured assessment of the drug–event relationship.
- WHO–UMC is the causality framework used by PvPI.
- The six WHO–UMC categories range from Certain to Unassessable/Unclassifiable.
- Temporal relationship, alternative causes, dechallenge, rechallenge and pharmacological plausibility are central considerations.
- Most spontaneous ADR reports are not “certain”; uncertainty is an inherent part of pharmacovigilance.
- Naranjo is a separate point-based tool and should not be confused with WHO–UMC.
- Causality, severity, seriousness and expectedness are distinct concepts.

17. References & Verification

1. World Health Organization / Uppsala Monitoring Centre. The use of the WHO–UMC system for standardised case causality assessment.

2. Pharmacovigilance Programme of India (PvPI), Indian Pharmacopoeia Commission. Toolkit — WHO–UMC causality assessment scale.

3. Naranjo CA, Busto U, Sellers EM, et al. A method for estimating the probability of adverse drug reactions. Clinical Pharmacology & Therapeutics. 1981;30:239–245. DOI: 10.1038/clpt.1981.154.

4. NCBI Bookshelf / LiverTox. Naranjo ADR Probability Scale.

Video reference: https://youtu.be/6gH4AWnwTDI

Prepared as professional study notes from the requested video topic and verified against authoritative pharmacovigilance sources. Always follow the current institutional/PvPI procedures for real-world case assessment.